

Month 2: Document Preprocessing, Chunking, and Lexical Search

This month focused on text cleaning, sliding-window chunking, parser testing, and baseline lexical indexing.

Week 1: Tokenization and Text Cleaning

- **Progress Summary:** Algorithms for text normalization and cleaning were implemented. An ingestion script was developed to remove extraneous characters, fix encoding inconsistencies, and prepare the raw HotpotQA Wikipedia corpus for tokenization.

Week 2: Sliding-Window Chunker Implementation

- **Progress Summary:** The sliding-window chunking utility (`pre_processing.py`) was implemented. A tokenization pipeline was built to partition cleaned text into segments of 256 tokens with a 32-token overlap, preserving context across chunk boundaries.

Week 3: Document Chunker Unit Testing

- **Progress Summary:** Unit tests were executed on the chunking module. Token counts and overlap segments were verified to ensure no textual content was lost. Boundary constraints were checked to ensure the parser handled short documents and empty files correctly.

Week 4: Lexical Search Indexing (BM25)

- **Progress Summary:** The lexical retrieval path was established using the BM25 algorithm. Text chunks were indexed locally to support keyword searches. Initial query tests were executed to verify the baseline recall of exact terms, names, and numbers.